

Part O: Reflection

Designing Verification Checkpoints to Mitigate Silent Errors in Agentic AI Pipelines

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46. Which part of converting the paper into LaTeX was easiest?

The easiest part was transferring the title, author information, abstract, keywords, section headings, and normal text into a structured LaTeX manuscript. This helped me understand how a research paper can be organized systematically using LaTeX.

47. Which part required the most manual effort?

Equation and figure conversion required the most manual attention. Mathematical notation, equation labels, TikZ libraries, figure sizing, and cross-references had to be checked carefully to ensure that the final document compiled correctly and preserved the intended meaning.

48. Which Prism feature was most useful and why?

The most useful Prism features were LaTeX conversion and compilation-error assistance. Prism helped convert manuscript content into LaTeX and identify problems involving the `\minimize` command, TikZ libraries, and figure width.

49. Did Prism introduce any technical, citation, numerical, or scientific errors?

During final verification, no unresolved citation, numerical, or scientific-content errors were found. The LaTeX conversion did produce three technical compilation/formatting problems, which were corrected manually: the undefined `\minimize` command, the missing TikZ geometry library, and an overfull figure width. These corrections did not change the scientific content or numerical results.

50. How much manual correction was required?

A moderate amount of manual correction was required, mainly for equations, TikZ figures, figure sizing, labels, cross-references, and compilation issues. The corrections were limited to LaTeX/document-formatting issues and did not require changing the scientific conclusions.

51. Did the same project compile in Overleaf without modification?

Yes. The exported LaTeX project was imported into Overleaf and compiled successfully without modification. The document structure, equations, table, figures, citations, references, and cross-references were preserved.

52. What precautions should researchers take when using AI for scientific writing?

Researchers should not accept AI-generated or AI-converted content without verification. They should compare the final manuscript with the original paper, check equations, numerical values, citations, references, terminology, figures, tables, and cross-references, and inspect compilation errors. AI should assist with document preparation, but the author should remain responsible for scientific accuracy and final verification.